

Executing Python Boto3 scripts on an EC2 instance using the attached IAM role

- There are two ways to execute Python Boto3 Scripts on an EC2 Instance
- They are:
 - Using the AWS Account Credentials
 - Using the Attached IAM Role
- Using the Attached IAM Role
 - The EC2 instance has an IAM role attached to it
 - That role has the required permissions (via policies).
 - Inside the instance, you run Boto3 scripts without needing to manually provide credentials or profile
 - Boto3 automatically retrieves temporary credentials from the IMDS

■ Steps to Execute Python Boto3 Scripts on an EC2 Instance using Attached IAM Role

- Step-1: Create or Use an Existing IAM Role
 - Go to IAM > Roles in AWS Console.
 - Create a role with:
 - Trusted entity: AWS Service
 - Use Case : EC2
 - Permission policies: Add whatever your script needs (e.g., AmazonS3ReadOnlyAccess, AmazonEC2FullAccess, etc.)
 - Name it something like EC2AutomationRole
- Step-2: Attach the IAM Role to Your EC2 Instance
 - Go to EC2 (if you don't have one already then please Launch Instance)
 - Select your instance → Actions > Security > Modify IAM Role
 - Attach the role you created (boto3AdminRole)
- Step-3: SSH into the EC2 Instance -> `ssh -i <your-key>.pem ec2-user@<your-ec2-public-ip>`
- Step-4: Install Boto3 (if not already installed)
- Step-5: Create and Run Your Boto3 Script
 - Note:
 - ❖ directly create a client as : `client = boto3.client(service_name='iam')`
 - (Or)
 - ❖ `session = boto3.Session()` and then `client = session.client(service_name='iam')`